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POWERING APOLLO

James E. Webb of NASA

W. Henry Lambright

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Introduction

On *May 25, 1961,* President John F. Kennedy addressed Congress and, in a now classic national challenge, declared his belief that the United States "should commit itself to achieving the goal, before this decade is out, of landing a man on the moon and returning him safely to the earth."[1]

A little more than eight years later, in an explosion of fire and cacophony of sound, the 3,817-ton Saturn rocket lifted the *Apollo 11* space capsule off the ground. Its destination, 238,000 miles away, was the moon. On board were three astronauts: Edwin Aldrin, Jr., Neil A. Armstrong, and Michael Collins.

Four days later, July 20, 1969, Armstrong climbed down the ladder of his landing craft, placed his foot on the moon, and took "one small step for [a] man, one giant leap for mankind."

On July 24, the small remnant of the once mighty space machine broke through the earth's atmosphere, ablaze from the friction of entry. Parachutes erupted, and the capsule gradually splashed down into the Pacific Ocean. The three astronauts, all safe, were soon aboard the recovery aircraft carrier. On a large screen at Mission Control in Houston, where the National Aeronautics and Space Administration (NASA) had been tracking the entry flight, the words of President Kennedy appeared, followed by "TASK ACCOMPLISHED, July 1969."[2]

In Washington, D.C., a private citizen named James Edwin Webb breathed a deep sigh of relief. His own personal mission to the moon was at last over. The national celebrations that followed *Apollo 11*'s triumphant success did not glorify Webb. He stayed largely on the sidelines, while others owned the spotlight, for he no longer held a position at NASA. But for most of NASA's journey to the moon, from 1961 to 1968, he had been the man in charge. As administrator of NASA in the 1960s, he had been the power behind Apollo. When he died on March 27, 1992, and was buried in Arlington National Cemetery, his special role at the helm of NASA was widely acknowledged. The *New York Times* obituary cited his reputation as "an extraordinary manager." The *Washington Post* wrote that he was "hard charging and immensely capable." Not a scientist, Webb was chosen as NASA administrator "for his management skills and political sophistication." Though he left NASA before Neil Armstrong walked on the moon, "he was universally credited with laying the groundwork for that epic event."[3]

It took approximately $24 billion to place a man on the moon (over $100 billion in current dollars), and the effort required the talents of twenty thousand industrial firms, two hundred university labs, and a team of four hundred thousand public and private workers.[4] Leading NASA in the 1960s was one of the greatest administrative and political challenges any individual has ever faced.

Who was James Webb? What did he do? How did he do it? What difference did he make as leader of NASA?

The Puzzle of Bureaucratic Leadership

Few questions are more important to ask or difficult to answer than those concerning governmental leadership. With our eyes focused on political leaders, few of us even know the names of the men and women who lead the large public organizations under them. They are part of the vast bureaucracy—that fourth branch of government about which little complimentary is said. Our color for bureaucracy is grey, our bureaucrats faceless. Many of us view *leadership* and *bureaucracy* as mutually exclusive terms, and a great deal of scholarly writing reinforces the conventional wisdom. In Herbert Kaufman's theory of organizational change, executive leadership plays little or no role. According to Kaufman, organizations that fail to adapt to shifts in their environment wither and die; organizations that prosper do so less because of leaders than of luck. In his extreme theory of environmental determinism, leaders have little impact on policy and programs.[5]

The other view, of course, is the "great man" theory, in which history is painted on canvas by outstanding individuals. Declares Thomas Carlyle, the nineteenth-century historian: "They were the leaders of men . . . the modelers, patterns, and in a wide sense creators, of whatsoever the general mass of men contrived to do or to attain."[6] In this view, organizations are extensions of their leaders.

Truth, as in most cases, lies somewhere in between. For the most part, individuals at the top of organizations likely have only marginal impact, especially if they happen to be there for but a brief time. But once in a while, when the context is right, an individual can make a difference in the success of an organization. And when that happens, the achievement is worth noting and attempting to understand, especially in U.S. public administration. For in the United States not only do great constraints prevent one from placing one's mark on a public organization but also a widespread belief holds that an individual bureaucrat should not have too much room in which to maneuver. After all, bureaucrats are not supposed to make policy. That is the task of elected officials in the United States, at least in theory.

Again, reality is often different. Ample evidence suggests that certain bureaucratic executives are major players in policy, and that an informal and unofficial reality requires them to play significant roles if American democracy is to function. Policy means little if it is not implemented, and the agencies and departments of government must lead this implementation. The way they lead is often deliberately obfuscated, however. Effective administrators in government often employ the strategy attributed to President Dwight D. Eisenhower—"hidden-hand leadership."[7]

Indeed a fourth branch of government, the U.S. bureaucracy consists of a multitude of agencies, departments, commissions, and government corporations. While exercising their authority under legislative and constitutional mandates, they inevitably have a degree of autonomy. Some individual agency executives use their discretion effectively, and some do not. Once in a while, an individual uses the position of public executive with such exceptional force that he or she is noticed—even remembered. As is J. Edgar Hoover, longtime chief of the Federal Bureau of Investigation, they are more often remembered for their abuse of power in office than for its use in the public interest.

Yet, if government is to work well, administrators must be willing and able to provide positive leadership. They must motivate followers, raise the performance level of their organizations, and accomplish the purposes for which their organizations were established and funded. Most leaders, able to maintain their organizations, are "transactional" leaders, in the words of

James MacGregor Burns. Those who create new possibilities, "transforming" leaders, change organizations and imbue them with values that last beyond their tenure.[8]

This administrative biography of James Edwin Webb, who played a "transforming" role in American public administration, tells the story of the man who built NASA in a few years from a new, small, vulnerable agency into an organization capable of taking America to the moon.

Webb's NASA has subsequently been cited as a model of "excellence in management of large-scale public enterprises." According to one observer, "Webb's tenure at NASA set the standard by which all subsequent administrators have been judged. He was called the quintessential manager."[9] Still, Webb's record at NASA was not perfect: he was administrator when three astronauts died in a fire that should not have happened; he left NASA with no firm post-Apollo goal; and he launched a mammoth effort to reform universities and regional economies that some regarded as ill-advised and that did not achieve the goals he set.

Nevertheless, NASA got to the moon on time and within the general costs projected, a monumental achievement by government and its private-sector partners. Project Apollo, the key NASA program, was one of the most impressive demonstrations of technological prowess in history. For most of it Webb was ultimately responsible. While Apollo had its critics then and now, it stands as a symbol of what the United States can do when it decides on a course and puts the necessary resources behind it.

Operating in the Cold War context of his times, Webb made the most of the opportunity his place as head of NASA presented him by influencing the Apollo decision, forcefully and skillfully seeing to its implementation, and even affecting the management of Apollo after he left. He pulled together the scientific, engineering, personnel, budgetary, political, and governmental efforts required. Apollo represented a Cold War technological victory, and he was the general who guided that victory.

Webb's administration of NASA culminated a series of major public assignments: director of the Bureau of the Budget (BOB) and under secretary of state in the administration of President Harry S. Truman. In every role, he tried to leave his mark, effectively in the cases of BOB and NASA, less so at the State Department, where his frustration illuminates the limits of administrative leadership in a particular context.

The son of a North Carolina county school superintendent, Webb worked his way through the University of North Carolina, was elected to Phi Beta Kappa, and graduated in 1928. After serving first a member of Congress and then a leading New Deal Washington lawyer, Webb joined Sperry Gyroscope Company in 1936, rising to vice-president in seven years. In 1946, President Truman chose him for director of the budget, then for under secretary of state from 1949 to 1952. During the Eisenhower years, Webb served as president of Republic Supply Company and assistant to the president of Kerr-McGee Oil Industries.

When President John F. Kennedy appointed him NASA administrator at age fifty-four, Webb looked and acted the successful man he was. Standing five feet nine inches, he dressed impeccably, was always well-groomed, and typified the midcentury executive who moved easily among corporate and governmental elites.[10] He had thick, wavy, dark-brown hair, "graying smartly, and combed straight back." Every strand seemed "nailed in."[11] Although stocky, he did not look overweight. Invariably polite, he had a disarming southern accent, but his square jaw matched the forcefulness that usually led him to dominate conversations. Said one NASA official, "Trying to make conversation with Jim Webb is like trying to drink out of a fire hydrant." "A man with a florid complexion and an intense manner that gave his blue eyes a permanently pleading, worried look," he was known for his "memorable, gesticulating, humorless volubility, with frequent interjections of a tic-like rhetorical question: Do you follow me?" His intensity was sometimes overwhelming. "If you ask [him] a question," said a top executive of one government agency, "you get fifty facts when two or three would do, but he has the knack of getting large numbers of people moving. I'm damned if I can understand how he does it."[12]

Perhaps it was the sheer energy of the man that made him such a memorable leader. Perhaps beyond the strong personality lay a set of management techniques he understood and practiced. Without a doubt, Webb believed in a Holy Grail of management, and that he had found it at NASA.[13]

Webb's Disputed Legacy

While not widely known to the general public, Webb is, among professionals in government and close students of government, a Washington legend. President Lyndon B. Johnson called him his best administrator, and bureaucratic veterans look to Webb for lessons in how to accomplish great deeds through public policy.[14] Accorded innumerable awards, testimonials, and honorary degrees (at least thirty-two from universities across the country), Webb had a special fund and lectureship for Excellence in Public Service established in his name. He has been featured in various lists of "giants" in management in the twentieth century and among the top "public entrepreneurs" the United States has produced.[15] A leading business management scholar, Leonard Sayles, called Webb a manager of "heroic" proportions:

> No one has ever adequately pointed out that, under Jim Webb, NASA was a magnificently managed program. As a society, we went from almost zero knowledge of man in space to a fully operational space program in less than ten years. I've seen a company take that long to build a new plant or get a new process working. Webb achieved an enormous accomplishment in organizational leadership. He contributed more than one can easily say to our national welfare. In the process, he showed himself to be an extraordinary human being. Yet the accomplishments of NASA are not associated in our minds with leadership or administrative ability, but with the courage of some superb astronauts and with the new technology of rockets and space capsules. Hardly ever, if ever, do we hear that this was management at its finest.[16]

Similar views of Webb abound. A reviewer of a book on space policy referred to him as an administrative "genius" who was able to build and control "NASA's vastly intricate structure, while never forgetting for a moment his need for full congressional support."[17] Webb had been called an administrative genius before—by Dean Acheson, whom he served as under secretary of state in a difficult, sometimes strained relationship.

If ever a man thought and talked about administrative leadership, especially in the public sector, it was James Webb. He read books about it, engaged scholars in debate on issues in the field, provided funds for university fellowships linking science, technology, and management, served as president of the American Society for Public Administration, wrote a book about the management of "large-scale endeavors," and helped found the National Academy of Public Administration to honor administrators as national academies honor scientists and engineers. Yet, some of the factors that made Webb so remarkable a public executive caused him also to have his detractors. For them he was a "combative bureaucrat who guarded his turf with canine ferocity,"[18] a "slippery, corny, insincere opportunist" whose championing of space was because it gave him "a bureaucratic empire." At the time of the Apollo fire, his stonewalling thoroughly exasperated the *New York Times* and such liberal Democratic critics as Senator Walter F. Mondale. The *Times* questioned Webb's "candor" and called for his ouster, as other media and congressional observers quipped that NASA stood for "never a straight answer."[19] His principles of large-scale management, celebrated by management scholars Leonard Sayles and Margaret Chandler, have been questioned by historian Walter McDougall, who labels Webb a "Big Operator" and a "technocrat."[20]

In his award-winning history of the space program, McDougall paints a rather unflattering picture of Webb as a man who saw NASA not just as an agency of government but as "a model of integration, a team effort drawing confidently on government, industry, and university," and Apollo not as a race with the Soviet Union but as an all-purpose engine mobilizing the country toward a range of larger ends. NASA, organization, technology—these were instruments for Webb.[21]

While McDougall speaks of Webb's surpassing ability and service-oriented motives, he is wary of state-funded and state-directed technological and social change, a view that others have shared. In the 1960s, admirers of Webb marveled at NASA's mobilization skills and agreed with him that NASA was a "wave of the future," an alternative to the "military-industrial complex," a dynamic, positive aggregation of public and private power that could serve in the public interest. If the nation could integrate government, industry, and universities to go to the moon, it could do the same for a host of other civilian problems. Then, as later, however, Webb's detractors said that NASA simply augmented the existing "contract state" and was on its way toward creating the administrative means to turn government into "Big Brother."[22]

Webb, therefore, represented a threat to critics of big government, because he exemplified the superbureaucrat, the man who makes the "megamachine" of government operate, seemingly oblivious to the dangers of concentrated power. That Webb himself wrote about the need to make such aggregated power accountable to elected officials made him no less worrisome to social critics of government-sponsored large-scale technology.

Was James Webb a "heroic figure" or a "big operator"? The question cannot be avoided because he was, in truth, something of both. To unravel his many aspects is to deal with the ambiguities and complexities of bureaucratic power in American democracy.

Webb thought of himself as an administrator, and he consciously sought to create in NASA a "perfect organization." But what virtually everyone else also saw in him was a skilled administrative politician, "supremely capable," at NASA particularly, in the art of "power brokering."[23] The

American people and the system of government they created are suspicious of power. The Constitution explicitly separates power among institutions to ensure slow, deliberate decision making. Checks and balances abound to make it difficult for any one person or group to wield too much influence. Yet to accomplish certain public purposes, power has to be focused. Institutions have to converge. Although political parties play a role in aggregating interests, slack party discipline and the geographical orientations of party members limit their influence. Administrators, responsible to the president and Congress, find they must build coalitions to be effective. Here lay Webb's true genius: his understanding of the relation of administration to power in Washington. To get America to the moon, he exercised governmental power in a way that provided NASA the resources and autonomy it needed. Whether his means were beneficial or harmful in the long run depends on one's view of both bureaucratic power and the Apollo program, but their effectiveness was undeniable.

Norton Long, in a classic statement, wrote that power "is the lifeblood of administration." He went on to say that

> its attainment, maintenance, increase, dissipation, and loss are subjects the practitioner and student can ill afford to neglect. Loss of realism and failure are almost certain consequences. This is not to deny that important parts of public administration are so deeply entrenched in the habits of the community, so firmly supported by the public, or so clearly necessary as to be able to take their power base for granted and concentrate on the purely professional side of their problems. But even these islands of the blessed are not immune from the plague of politics. . . . To stay healthy one needs to recognize that health is a fruit, not a birthright. Power is only one of the considerations that must be weighed in administration, but of all it is the most overlooked in theory and the most dangerous to overlook in practice.[24]

Whatever else can be said about him, Webb did not overlook power in his practice of public management. He built, in and through NASA, a power base of huge proportions. It is in part because he did so that NASA was successful, and it is also because he succeeded that he had adversaries. Power is a concept about which there is great ambivalence—especially bureaucratic power. Aware of this, Webb frequently noted that there was a "line" the responsible public executive had to walk.

The story of James Webb is as much a story of administrative power as of administrative accomplishment, because one made the other possible. Moreover, while Webb built power for himself and his agency, he also recognized its limits. Because he knew failure as well as success in his career, he sensed when power had been lost. At NASA, he sought to create a bureaucratic momentum that could last beyond his tenure.

In doing so, he defended the distinctive competence of his organization against the Defense Department and other agencies that challenged NASA's quest to lead in space policy. He pressed the disparate, competitive parts of NASA to work together in a more cohesive whole. He built coalitions involving Congress, president, media, and more. He fashioned an alliance embracing industry, universities, and others on whom NASA depended. He fused administration with science and technology. He used the financial resources of NASA to reward and punish. He employed rhetoric and other intangible resources to accomplish his ends. He was both administrator *and* politician.

Tom Wolfe, author of *The Right Stuff*, called Webb one of the "ablest and most distinguished of the off-the-ballot politicians," "a valuable breed well known in Washington." He described Webb as "the sort of man of whom a congressman or senator was likely to say: 'He speaks my language.'" But more than a politician, Wolfe noted, Webb was an administrator "known as a man who could make bureaucracies run."[25] Armed with a national goal set by the president, as well as a personal agenda set by himself, in order to succeed he used every legal and legitimate means at his disposal—with guile, against influential opponents. The congressional consensus behind Apollo lasted barely two years. By 1963, NASA's requested budget was heatedly debated, and while NASA received more than in 1962, it obtained far less than Webb had sought. In the mid-1960s, the country turned to the Great Society, which substantially expanded social programs and federal spending, and by the late 1960s, the overriding priority, Vietnam, was tearing the country apart. The wind shifted steadily against Webb as priorities changed and rose sharply against him in 1967, when the Apollo fire killed three astronauts. The way Webb handled that crisis highlighted his strengths and weaknesses. In the view of some, he allowed his end to justify his means, violating proper administrative accountability in a democracy. Webb's conduct then in defense of NASA helped make him the issue.

His reputation diminished by the experience, Webb survived the Apollo fire and its aftermath, drew even more power to himself within the agency, and propelled NASA toward a full recovery where Apollo was concerned. But when he could not sell a post-Apollo program, he largely abandoned the task, casting aside most of the items on his personal agenda, such as using universities to strengthen regional economies and moving the United States toward what McDougall called a "Space Age America." In the end, Webb used his diminishing political capital to give NASA one last push toward the moon, finally sacrificing his own position on the altar of Apollo.

As astronauts walked the moon, Webb proclaimed to all who would listen that Apollo's real achievement lay in demonstrating that a democratic nation could outmanage an authoritarian state. He said that if the United States could go to the moon, it could solve its other public problems. Few listened. Richard M. Nixon, president in 1969 when the moon landing took place, ushered in a lengthy conservative era. Rather than an engine for problem solving, government was depicted as part of the problem. Even Democrats—among them President Jimmy Carter—cast aspersions against government, especially government bureaucracy. Big technology, which Webb embraced, was also suspect in the wake of Vietnam, a counterculture movement, and the rise of environmentalism. The longstanding partnership between federal officials and scientific and technical experts, forged in World War II, was shattered. Apollo seemed the end of an era rather than the prototype for a future civilian science and technology relationship.

Webb continued a productive life after NASA, but his message was increasingly out of touch with the times. He died at eighty-five, a resident of Washington, D.C. Much honored, indeed idolized by many in NASA who had served under him, Webb was nevertheless frustrated. He believed the nation he loved and served had found a new frontier in Apollo, not only in space but in government management, but had stepped back from the opportunity and was the poorer for it.

Despite his leadership of one of the most extraordinary technological achievements in history, Webb is not well known. Avoiding publicity was a function of his personal style and administrative strategy. He understood the importance of sharing credit even in Washington, where credit is sought often for its own sake. A creator of the cult of the organization, not of the personality, Webb was more interested in NASA's aggrandizement than his own, although he certainly was intensely ambitious, conscious of his own personal power, and protective of his status. In many ways his career is a case study in what it takes to achieve administrative success in the United States—to be a master of the fourth branch of government.

1

The Making of a Public Executive

James Webb owed his achievements in government considerably to his North Carolina roots and his stable family, from which he reaped strong values, a positive outlook, and the desire to succeed. Born on October 7, 1906, in Tally Ho, North Carolina, he had an older sister and would be joined later by another sister and two brothers. Their father, John Frederick (J.F.) Webb, and mother, Sarah Edwin Gorham Webb, were college graduates in an era when that was the exception for most Americans. Both native North Carolinians, Webb's parents were education oriented and instilled in Webb a love of learning.[1]

An educator, J.F. was appointed education superintendent for rural Granville County, North Carolina, one year after Webb's birth, and the family moved to Oxford, county seat of Granville, a town of four thousand.

Quiet and cerebral at home, J.F. liked to read and was fluent in Latin and Greek. Active and dedicated in his profession, he was zealous about raising the quality of schools.[2] The progressive spirit moved the nation then, its emphasis being that society could be improved and that government and education should be forces for change. In the south, North Carolina "led the way by generating public enthusiasm for education, pumping large additional funds into schools, lengthening the school term, and increasing enrollments."[3]

As a county school superintendent, J.F. was caught up in this progressive mood. Having attended and later taught at a private school of considerable note in Bell Buckle, Tennessee, run by kinsmen W. R. "Sawney" Webb and John Webb, he wanted to bring the high standards he saw there to the

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LAUNCHING A STRONGER NASA

President Kennedy's decision to go to the moon changed a great deal for James Webb. It set a national goal with, upon the ensuing congressional endorsement, an initial political consensus. It was a national strategy within which Webb could perform as master tactician. But the NASA administrator was not sanguine about the honeymoon lasting the full decade. He knew his time was limited. While others used the word *commitment* in association with the president's decision, Webb did not. "I felt [Kennedy] had given us the authority to start," he recalled later. "It was up to [us] to go as far and as fast as we could, and bid for his support by doing a good job."[1]

Enlarging Apollo's Support

For Webb, the internal actions of the administrator were governed by the external circumstances, and vice versa. He had multiple constituencies, beginning with the president. The most important fact about Kennedy's decision, for Webb, was that it linked the president's prestige, possible political fate, and place in history with his own. Webb had wanted a direct relation with Kennedy from the outset but had found himself one of a group of officials dealing with him. Now, Webb was Kennedy's prime instrument for getting to the moon. He depended on Webb to make him look good, and that gave Webb power vis-à-vis Kennedy.

It also settled a matter important to both men—the role of Lyndon Johnson in space policy. Webb was perfectly willing to deal with Johnson and let him take credit and think he had power, but he did not want Johnson running NASA. When Webb talked with the president about Johnson and reporting relationships in general, Kennedy said he should deal directly with him and the budget director. Webb suggested a way the president could keep Johnson occupied on important matters, yet make sure he did not become a problem to them: make clear to the vice president that Kennedy would set the agenda of the Space Council. No one would lose face. Kennedy agreed, and Johnson and the Space Council were soon completely occupied with communication satellite policy.[2] Also, Senator Robert Kerr, chair of the Senate Space Committee, "performed a great service for me," Webb recalled. "He told Johnson this story about my independence [the Webb-Kerr relation in Oklahoma], how I wouldn't be kicked around, which meant Johnson never tried."[3]

Apollo also raised Webb's bargaining capacity with Congress. He felt he "had as much capability and power to make trouble for an individual member of Congress as he had to make trouble for me."[4] Armed with a national mission, Webb was now in charge of the most visible front of the technological Cold War with the USSR. Moreover, he would have lots of money to dispense. How and where that money was spent mattered to Congress. As Webb put it, the legislators "were interested in selling their influence with the space program for other kinds of benefits they could get in their districts and committees." Webb had to tell them that he was "going to get this job done." To do that he had to "hold enough power within [his] own right personally to make sure it would be done," which meant excluding other influential people from decisions.[5]

Above all, he had to succeed—and to appear successful.[6] He had to convince influential legislators (as well as others) that he "knew what [he] was doing and that [he] could be a winner. They would not support a loser." The most significant man in the Senate for Webb was Kerr, who was, said Webb, a "protector of NASA. But he was also a pressure point. His style was to cooperate with people who cooperated with him. You had to convince Kerr that you could help his image and you could help him with the White House. You had to offer him something. And, if you could not offer him what he wanted, you had to offer him something else instead."[7] Shortly after the May 25 presidential announcement, Webb made it a point to go to Oklahoma to help Kerr stage a meeting on space policy with various local and regional leaders so the folks back home could see that he was a major figure in Washington in this glamorous new field.

In the House, the most important person for Webb was Albert Thomas, chair of the Appropriations Subcommittee that set NASA's budget and a Texan who also expected an "exchange" for his political support. In dealing with Thomas and others in Congress, Webb studied the incentives that made them move this way or that on space. He also knew that he would be asking for huge sums and that he could help make his case that every penny was needed if he did not appear to be living luxuriously as administrator. Instead of using the limousine to which his position entitled him, he drove in an old Checker cab, painted black. "It's the little things that can get you into trouble in Washington," he told his friends. As he prepared to build a multibillion-dollar agency, he acted—at least around some in Congress—the frugal country boy.

Politicians were only one of Webb's constituencies. Although he had said he would march under the banner of management innovation, he also emphasized continuity in his first months at NASA so that he could absorb information from his organization. "To be innovative," he said, "you have to start somewhere. What I needed to know from NASA was what had worked in the past. I needed to know what NASA people knew. But I brought to NASA a whole bunch of theory and doctrine."[8] The emphasis on continuity shifted after the Apollo decision to one of financial and organizational change.

Webb's immediate staff was small. He began with his personal secretary and executive assistant and gradually expanded. His executive assistant, R. P. "Rip" Young, became Webb's extra eyes and ears; as the job enlarged, Young brought in other people to augment the flow of information. They became a secretariat, but a much smaller and less threatening version than Webb had created at the State Department, intended to keep track of information coming into the administrator's office and make sure it was channeled to those who could benefit from it. Webb wanted each member of the triad to be informed of whatever came in to any one of the three. He also wanted the secretariat to keep after the program heads to get them to make sure information came up to the policy leaders that should come up, in return for information that came down.[9]

Working closely with the leadership triad on all substantive decisions, Webb met regularly and as necessary with Hugh Dryden and Robert Seamans, each of whom had his own area of responsibility.[10] Webb's outside political role was reflected in those within headquarters with whom he worked most closely—Paul G. Dembling, who was in charge of the Office of Legislative Affairs, and Hiden T. Cox (later Julian Scheer), in charge of Public Information. These were Webb's prime points of contact between NASA and Congress, and NASA and the media.[11]

This topside emphasis on congressional and public relations was new to NASA, as was Webb's use of a succession of consultants. His predecessor, T. Keith Glennan, had used consultants in a formal way, for management decisions and organizational planning. Webb eschewed such uses. Perhaps arrogantly, he thought he knew more about management than any consultant. Those he used gave him ideas about program substance or helped him keep tabs on what was going on in his organization. The latter group he called "scouts."[12] His principal scout was Arthur Raymond, retired top executive from Douglas Aircraft, who joined NASA in 1961 and stayed throughout Webb's tenure, roaming the NASA system and providing him "early warnings" on developing problems.[13]

Webb's administrative style combined the formal and informal. At the apex of a pyramid, he respected the chain of command where appropriate but ignored it at other times. As well as part of the triad, informally he was at the center of a wheel whose spokes carried information both ways; he conversed on the phone with program directors and center directors, making it clear they could see him as needed. Webb proclaimed to all he encountered that he would "use the best administrative theory and doctrine we can get." He realized that neither Congress nor the scientists and engineers in NASA had any idea what he was talking about, but he said it anyway. What Congress wanted, said Webb, was "a man they could trust who could get the job done and would make them look good." "Management innovation" and "orderliness in administration" were his flags. He wanted those who observed NASA from outside to have confidence in the agency rather than to look "askance" at it.[14]

In dealing with both his internal and external constituents, Webb moved quickly but with forethought. Government involved separation of powers and checks and balances, and one of the powers was bureaucracy. It provided administrative competence, organizational efficiency. As a fourth branch of government, it had to be accountable, although this did not necessarily mean Webb was going to clear everything he did with others. He took the view that people should judge him by his results rather than by his plans. He did not want senior officers telling him what to do. "I figured it was my responsibility to keep going until the President stopped me," he said, "and to have a program he would want to approve rather than disapprove."[15]

While Webb told his political superiors what he thought they needed to know, he strove for the discretion he regarded as necessary to succeed. This meant he had to satisfy a variety of internal and external constituents that they could realize their interests through him. At the same time, he had to take risks: do what he thought was necessary, justify his actions afterward, and personally absorb the political heat. In the year or so following Kennedy's Apollo speech, Webb made rapid-fire decisions, taking advantage of his honeymoon with political superiors to propel NASA on its trajectory toward the moon, as well as to initiate other items on his agenda.

An early example involved the contract for the guidance system, one of the first and most important Apollo contracts. Although Webb had set up an elaborate Source Evaluation Board procedure to insulate major procurement decisions from external political pressures and to emphasize considerations of technical merit, he was not about to let procedures hold up progress. In this case, he was certain that the best person in the world to design a navigation system to the moon was Stark Draper, leader of MIT's Instrumentation Laboratory. As a former Draper associate, Seamans also saw him as a unique talent, unsurpassed. He had done extraordinary work on guidance systems for the military. Designing the Apollo navigation system, a completely original task, could not be allowed to fail. Before making the final decision, Webb asked Draper if he could really develop a space navigation system that could get a man to the moon and back safely. Draper said he was so confident that he would be willing to make the voyage himself, if Webb would guarantee the propulsion system. That was enough for Webb. In August, NASA awarded a sole-source contract for the Apollo navigation system to MIT. Industry complained, but the process moved forward.[16]

The Apollo decision also meant a vast expansion in NASA facilities. In August, NASA acquired 125 square miles of land on Merritt Island (Cape Canaveral), Florida, for a spaceport, which would eventually grow into a new NASA center headed by Kurt Debus. The next month, NASA selected Michoud, Louisiana, as the production facility for Saturn rockets, and debated the location of the biggest prize of all—the Manned Spacecraft Center (MSC), whose mission would be to develop the spacecraft and create the complex of technical facilities for spacecraft research and development, astronaut training, and flight operations.

When Webb decided to locate the MSC in Texas, he crafted a memo to the president that set forth the technical criteria and explained why a site outside Houston was chosen. Kenneth O'Donnell, presidential appointments secretary and a key Kennedy political operative who favored a Massachusetts site, recalled that "Webb made an excellent argument. The President can read. He said, 'It's a good decision. Let's go through with it.'"[17] Webb told Robert Gilruth, chief of the Space Task Group at the Langley Center in Virginia, who would be MSC's director, that he and many others would be moving to Texas. He mentioned the technical factors in the decision, but, as Gilruth recalled, also asked, "What did Harry Byrd [senator from Virginia] ever do for you, Bob?" When Gilruth replied, "Nothing," Webb responded, "That's what I mean. We've got to get the power. We've got to get the money, or we can't do this program. And the first thing, we got to move to Texas. Texas is a good place for you to operate. It's in the center of the country. You're on salt water. And it happens to be the home of [Albert Thomas] the man who is the controller of the money."[18]

The contract with MIT, the first of many important contracts NASA let in 1961 for Apollo, was unusual in going sole source to a university. Most of NASA's work would go on a competitive basis to industry, with the bulk of the major contract decisions for Apollo made between September and December. For example, each of the three stages to the Saturn rocket went to a separate firm: in September, the second stage went to North American Aviation; in December, the first stage went to Boeing and the third stage to Douglas Aircraft. In between came the most controversial decision: the contract for the command and service module, the Apollo spacecraft.

The Source Evaluation Board, made up of technical experts drawn from within NASA, in November recommended the Martin Company as "the outstanding source for the Apollo prime contractor." Webb, Dryden, and Seamans, as was their policy, questioned the board carefully, then met in Webb's office for the final decision. When Gilruth and others who would be directing spacecraft development were brought in and asked to comment, they protested the board recommendation, indicating that they spoke also for a number of astronauts; all felt that Martin knew how to build missiles, not machines that flew men. Their choice, rated second by the Source Evaluation Board, was North American, which was able to build "flying machines" and had demonstrated exceptional work on experimental aircraft.[19]

Webb said he wanted to sleep on it over the weekend. His dilemma was that NASA was divided, that his policy was to spread contracts around to diversify the agency's industrial and political base and enhance competition, and that his scouts told him North American was already "strained to the limit" with the Saturn second stage. Should NASA's largest single contract award go to the contractor that already had the most NASA work? Webb had been pressured by Fred Black, North American lobbyist, who had promised North American would expand operations in Oklahoma, thereby helping Senator Kerr—and thus Webb. Webb placed a call to his former

Truman administration associate Robert Lovett, his predecessor as under secretary of state, now a member of the North American board. Could North American assume another huge and technically demanding job? Lovett told him he could "count on this company." On Monday morning Webb told his associates he was going with North American.[20]

Choosing a top technical manager to direct Apollo was another crucial decision, one of the most important Webb made in 1961. Throughout the spring and summer, Webb, Dryden, and Seamans discussed possibilities among themselves and with contacts outside NASA. Two candidates emerged within NASA, Abe Silverstein, head of the Office of Space Flight, and Wernher von Braun, director of the Marshall Space Flight Center (MSFC) in Huntsville, Alabama. Unfortunately, the potential tension between the two men, owing to Silverstein's Jewish heritage and von Braun's service to Hitler in World War II, was a matter that could not be avoided, at least in private conversations. But which should head manned space? Von Braun was summarily rejected by Dryden, who told Webb and Seamans that they could select him, if they insisted, but he would submit his resignation the same day. Webb decided to talk to Silverstein. The conversation did not go well. As Webb saw it, Silverstein wanted to be a czar with virtually complete autonomy, especially in regard to Seamans. Webb looked elsewhere within government. The most successful recent large-scale project was the navy's Polaris, and the deputy to Admiral William Raborn, head of Polaris, had been Captain Levering Smith. The NASA leaders had a good discussion with Smith, and it looked as though he was going to join NASA as Apollo director. But as soon as the navy found out, it promoted Smith to admiral and gave him an assignment he could not refuse.[21]

The NASA triad searched further that fall, outside government, and Seamans came up with D. Brainerd Holmes, of Radio Corporation of America, with whom he had worked in the past and who, at age forty, was considered one of the top technical executives in the country. Holmes had been directing the design and construction of the Ballistic Missiles Early Warning System, which involved massive installations in northern Alaska, Greenland, and Scotland. Executives in industry whose judgments Webb trusted spoke highly of him. Webb met Holmes and decided that he had qualifications that made him a good candidate. By November 1, Holmes was at NASA, taking a huge cut in salary in his move from the private to the public sector as director of the new Office of Manned Space Flight (OMSF).

Reorganizing NASA

At the same time, Webb had reorganized NASA for its new responsibilities, especially Apollo, by creating four program offices: Advanced Research and Technology, Space Science, Applications, and Manned Space Flight. The Office of Advanced Research and Technology essentially encompassed the former NACA enterprise (i.e., the older aeronautics centers). Webb had carved the space component of NASA—mostly under Silverstein's Office of Space Flight—into three rapidly growing entities. That Science and Applications now enjoyed an organizational status equal to that of Manned Space Flight was a deliberate move by Webb to check and balance the inevitable power its funding would give OMSF and to emphasize his view that NASA was more than Apollo. Seaman's authority under the reorganization was greatly enhanced: all program *and* center directors reported to him, and he controlled budgets.[22] When the reorganization was publicly announced, its justification came in the jargon of administrative rationality. At issue in fact was power. Webb had pulled power upward from the centers to headquarters, especially to the triad, to counter centrifugal forces operating at midlevel and to provide a measure of balance among NASA programs. He wanted internal control at this point, when so much was getting started, including a third manned program.

When Holmes joined NASA, he had two manned programs to run—Mercury, well under way, and Apollo, still being planned. As of December 7, he had a third, not even conceived at the time of Kennedy's decision—Gemini. The reasons for Gemini were primarily technical, but they also served Webb's need to consider the political base for Apollo.

Planning for Apollo indicated that "the leap from Mercury to Apollo was prodigious." Apollo would need three astronauts, instead of Mercury's one, to maneuver the large spacecraft; furthermore, one gigantic spaceship/rocket (the proposed Nova) capable of going to and returning from the moon could not be developed in time to meet the president's deadline. This "direct ascent" approach would have to give way to a "rendezvous" method: a smaller (but still large) system would be launched and somehow reassembled outside the earth's gravity for the remainder of the voyage. For the lunar mission to be successful, Apollo's astronauts would need to learn how to use new radars and computers to follow and link up equipment, or "dock."

For Gilruth, it was too risky for NASA to wait the projected four years between the end of Mercury and the first flight tests of the Apollo system. A program had to be devised to give the astronauts experience in all aspects of lunar flight, especially rendezvous. The solution was Gemini, which would use a two-man capsule.[23] In announcing the Gemini decision in December 1961, NASA officially proclaimed it "the training program for Apollo," although Webb privately regarded it as "insurance against a possible Apollo failure." "Mercury would not have led us very far," he noted. "With Gemini, we would have learned a great deal about spaceflight and its capability. If we had an insuperable obstacle and had to stop Apollo, if our equipment wouldn't work or it is too difficult a job, if we really didn't see how to overcome some difficulty in getting to the moon, we would have still done the next most important thing."[24]

Also, Webb appreciated the political dividends. Gemini would keep NASA visible during the hiatus between Mercury and the first Apollo flights, the perfect research and development laboratory for NASA's technologists—an evolutionary machine from which they could learn in moving toward Apollo. McDonnell, the contractor for Mercury, would build it. Webb explained the rationale of Gemini to the president and Congress, who went along with the decision.[25]

While Webb reshaped NASA for Apollo, he also launched what would be known inside and outside NASA as *his* university program. Scientific consultants told him in summer 1961 that the country had scientific needs but questioned NASA's "authority" to supply such requirements. Webb's "eyes lit up," John Simpson, a University of Chicago space scientist, recalled. He said, "That's my job to worry about that. I'll see to it that our charter let's us do what we need to do."[26] Webb's idea of needs went beyond only what was good for science. He wrote Lee DuBridge, president of Caltech, that billions would be channeled through NASA (much of it to the West Coast) over the coming decade to advance science and technology at the most rapid rate, and "to feed it back into our national economy and the fabric of our national life." He sought to interest DuBridge in helping him find a way by which NASA-derived technology could energize urban development, water resources, energy, communications, management, and life sciences. "I know that the lunar objective causes some problems to you," but it would prove its worth "even if we never make the lunar landing."[27] DuBridge and other university presidents to whom Webb communicated no doubt wondered what the exuberant NASA administrator was up to. What did such earthbound issues as urban development, water resources, and energy have to do with going to the moon? Was Webb running NASA or the whole government? And what did universities have to do with all this?

When Webb spoke in general terms to a number of key legislators about a new "university program," few paid much attention or raised objections. They all had universities in their states and districts. If Webb wanted to expand his work with universities, why not? Of course, they wanted to make sure there was sufficient geographical spread in the program. Webb reassured them. This would not be an elitist program.

NASA's revised budget included new authority for NASA to provide facilities to universities, thus remedying one worry Webb's scientific consultants had raised. When the November NASA reorganization was announced, an existing Office of Grants and Research Contracts was moved under Homer Newell, head of the new Office of Space Science. Webb told Newell he would reprogram $12 million from the budget to get the Sustaining University Program (SUP) started this fiscal year and would see about enlarging the program in the next.

Webb knew that President Kennedy wanted to increase funding for science education, including higher education, that Jerome Wiesner had told him of a need for thousands of new scientists and engineers to be produced in the 1960s, and that Kennedy had been rebuffed by Congress in his attempt to increase the number of students trained in science, engineering, and mathematics through other agencies. Webb asked him to consider NASA as an alternative vehicle, noting that NASA was being criticized by Wiesner and others for taking too many scientists and engineers away from more "worthwhile" endeavors. Well, Webb said, he would "fill the tank from which we were drawing." He asked Kennedy to back him up when the budget director challenged him on this, and, as Webb saw it, the president gave him the "go ahead."[28]

In late November, Webb and his budget assistant met with Budget Director David Bell to discuss the next year's budget. Alerted by his staff that Webb had reprogrammed money in 1961 to get into the university business and was proposing to extend its domain in 1962 to the tune of $30 million, Bell asked Webb what this was all about. Webb defended the effort, and when he finished told his assistant to "bury" the university funds within the general NASA budget request. No line item for facility grants, institutional research grants, or training grants (fellowships) would appear, in contrast to the original NASA proposal sent to the Budget Bureau. SUP would be made less visible to Congress and would be funded again through reprogrammed funds. Webb said he had Kennedy's sanction for the program, which Bell could check with the president himself. Although Bell did not debate Webb, he asked for a letter fully explaining his program.[29]

Neutralizing Opposition

The Apollo honeymoon ended in 1962 as challenges to Webb's decisions began. During the first half of the year a spirited technical debate took place within NASA over whether to rendezvous near the earth or in orbit around the moon. Aware of the debate, Webb chose not to micromanage but to make sure technological choices benefited by processes giving all sides a chance to have their say, recognizing this as one of those times when waiting a bit longer would produce a better decision. His desire for consensus also reflected the fact that the technical debate on mode had gradually taken on an institutional complexion. Slowly but steadily Houston solidified around Lunar Orbit Rendezvous (LOR). Huntsville, led by von Braun, favored Earth Orbit Rendezvous (EOR). Because NASA could only go to the moon with its two Apollo development centers in tandem, with Houston responsible for the spaceship and Huntsville the rocket, Webb allowed enormous amounts of NASA time to be poured into the mode issue, a choice generally viewed as superseded in importance only by the president's lunar decision itself. Finally, von Braun announced that Marshall was going to go with LOR, which he felt offered "the highest confidence factors of successful accomplishment within this decade."[30] His shift essentially ended the debate within NASA.

Pleased with the process, Webb accepted the judgment of his organization only to discover that the president's science adviser, Jerome Wiesner, felt LOR was "the worst mistake in the world," one that was "risking these guys like mad." Throughout the summer and into the fall, NASA and Wiesner's office debated the technological choice. President Kennedy, Wiesner, and various dignitaries toured NASA facilities in September. At Huntsville, von Braun showed them a mock-up of the first stage of the moon rocket, the *Saturn 5*. "I understand you and Jerry disagree about the right way to go to the moon," the president said. "Yes, that's true," said von Braun. "We were having an intelligent discussion," Wiesner later recalled. "I was starting to tell Kennedy why I thought they were wrong when Jim Webb came up, saw us talking, thought we were arguing, and began hammering away at me for being on the wrong side of the issue. And then I began to argue with Webb." The vociferous dispute occurred in front of the president and the media accompanying him. On the way back to the president's plane, Sir Solly Zuckerman, science adviser to the British prime minister, who was part of the entourage, asked Kennedy how the dispute would come out in the end. "Jerry's going to lose, it's obvious," said Kennedy. "Why?" the Englishman inquired. "Webb's got all the money," Kennedy said, "and Jerry's only got me."[31]

In October, Webb informed Wiesner that NASA had studied the mode decision long enough and was going ahead. This would mean award of a contract to Grumman Engineering Corporation to build a small lunar module that would detach from the mother ship circling the moon, land on the moon, and then return to the mother ship. It would be up to Wiesner to stop him. In early November, Webb and Wiesner met in a tense confrontation before Kennedy in the White House, where Webb cast the issue in terms of who was in charge of getting to the moon. "Mr. Webb," the president said, "you're running NASA—you make the decision."[32] On November 7, Webb announced that the LOR program would go forward.

Webb fought internal as well as external battles during 1962–63, at times simultaneously, reinforcing his authority within as well as outside the organization. One of his most difficult internal problems concerned the original seven Mercury astronauts. As celebrities, they had a constituency of their own that included the president. They had signed contracts with *Life* magazine (prior to Webb's appointment) allowing them to supplement their government incomes with "exclusive" stories. In May word seeped back to Webb that the astronauts might be overstepping the bounds of propriety. They were being offered free homes by real estate interests in Houston, and some of them had no intention of living in the houses but intended to sell them and pocket the money. Webb anticipated a major threat to NASA's image unless he reigned in the astronauts, fast. When Kennedy heard about the venture, and that some media people were trying to link the White House to the astronauts' financial rewards, he called Webb, saying he could not understand how the White House had been dragged into the affair. Webb said he did not know how either, "but I can tell you how to get out of it. Just tell [the media] the Administrator of NASA is handling that."[33] Webb summoned the seven astronauts and Gilruth, their nominal superior at Houston, to Washington. Shortly thereafter, word went out that the astronauts had refused the gift of homes.[34]

The Mercury astronauts, however, had one more challenge for Webb. After the Gordon Cooper flight brought the Mercury series to a close in May 1963, the seven astronauts pressed for "one more" Mercury flight in the gap pending the first Gemini flight, not scheduled for some time. Webb told them that it was time to turn NASA's attention to the next step in going to the moon—Gemini. When the astronauts informed Webb they were going to take their case to the president, he told them to go ahead. He soon got a call from Kennedy. "You know who's going to make this decision, don't you?" he asked. Webb replied, "I think I do," and the president said, "You're going to make it." Webb then sent Seamans down to Houston to publicly state—in front of the astronauts—that Mercury was over, and it was on to Gemini.[35]

Far more serious than the astronauts as an internal test of Webb's authority was his relation to the man he had chosen to lead the Apollo project, D. Brainerd Holmes. Although Webb presided over the four divisions of NASA, Holmes's division, the Office of Manned Space Flight, controlled at least three-quarters of the budget, with Mercury, Gemini, and Apollo under its aegis, as well as by far the most people. Aggressive, determined, and immensely capable, Holmes expected utmost loyalty and did not mince words. Barely two weeks on the job, in November 1961, he stormed into Webb's office and demanded independence from Seamans. Webb refused.[36]

Webb and Holmes operated on different wavelengths. Holmes came out of meetings with Webb, saying, "I can't understand that man at all."[37] He never stopped pushing for more power and more resources and, although he was not necessarily wrong in his assessments of what was good for Apollo, his style grated on Webb. In 1962, NASA headquarters moved into two new buildings, one on and one near Independence Avenue, Holmes in one building, Webb in the other. The separation not only reflected the scale of Holmes's operation but also symbolized the distance growing between the two men.

Holmes, meanwhile, was becoming famous as the Apollo "czar," a term Webb hated, and *Time* magazine featured him on its cover August 10, 1962.[38] Holmes consciously cultivated Congress, much as Webb did. The NASA administrator tolerated the actions of his OMSF director, aware that he could not get to the moon without a hard-driving executive behind Apollo. His basic philosophy about strong and difficult people was that you had to take the bitter with the better.

By late summer Holmes felt the Apollo schedule was slipping—possibly four to six months behind schedule—and he determined he needed $400 million more than the original estimates to catch up. Since Congress was still in session, he asked Webb to request a supplemental appropriation for the project, funded at $2.2 billion for fiscal year 1963. Webb said he could not go back to Congress, a move that would harm his own credibility with legislators. Well then, said Holmes, why don't you simply transfer the money from these less important, nonmanned programs? Webb was even more adamant about cannibalizing the science, applications, and SUP efforts to get the $400 million. Holmes was angry. He believed Webb was placing Apollo—and Holmes's own chances of succeeding—in jeopardy.[39] Failing to make his case inside, Holmes secretly began talking with supporters in Congress and with the media, unaware that Webb's contacts kept him apprised of these activities.

The issue came to a head when Holmes sought an interview with *Time* magazine in which he discussed the dispute over priorities. In an advance copy of the story made available to a member of the Webb staff, *Time*'s editors had excised a quote from Holmes: "The major stumbling block of getting to the moon is James E. Webb. He won't fight for our program." The editors felt that Holmes had let the original cover story "go to his head," and that he was "out to get Webb."[40]

The *Time* article came out on November 23 depicting Apollo's alleged money problems, schedule slippages, and the disagreement between Holmes and Webb. Contrasting Webb's view that Apollo was only part of the space program with Holmes's perspective that "it's the top priority program within NASA," the article also mentioned that Holmes blamed Webb for letting the Apollo program lag four to six months behind schedule.[41] The story triggered attention in the *New York Times* and elsewhere.

Soon afterward, Kennedy called a meeting at the White House, where Webb and Holmes verbally fought it out in front of the president. Webb said he could not take responsibility for a space program that was not "balanced," waxing at length on the various facets of the space program and how all were critical to the country. When he was finished, Kennedy expressed his concern that he and his NASA administrator might not be in accord on Apollo's priority. He said that the manned lunar landing was the most important U.S. objective. Webb said, no, the objective was to be "preeminent in space." After Webb stated his position again, at Kennedy's request, the president said he was "still not sure" they saw "eye-to-eye." He asked Webb for his position in writing so he could be certain if they were working together or not.[42]

In a nine-page response, Webb wrote:

> The objective of our national space program is to become preeminent in all important aspects of this endeavor and to conduct the program in such a manner that our emerging scientific, technological, and operational competence in space is clearly evident. . . .
>
> Consequently, the manned lunar program provides currently a national focus for the development of national capability in space, and,

> in addition, will provide a clear demonstration to the world of our accomplishment in space. The program is the largest single effort within NASA, constituting three-fourths of our budget, and is being executed with extreme urgency. All major activities of NASA, both in Headquarters and in the field, are involved in this effort, either partially or full time.

While manned space had the "highest national priority," he went on, it would "not by itself create the preeminent position we seek." He described how the other space activities contributed to preeminence, and the importance of a "well-balanced space program." He would not recommend taking funds from these other areas for the manned lunar landing. Finally, he pointed out that he was discussing with BOB a NASA budget in FY 1964 that would increase the $3.7 billion appropriated in FY 1963 to $6.2 billion. Webb said he would rather wait for the regular budget-approval process with Congress, but that if the president determined that a supplemental appropriation for the current year was the way to go, he and his colleagues at NASA would do their best to present the case on Capitol Hill.[43]

Kennedy thought Webb was going too far beyond Apollo with some of his "preeminence" interests and in his heart of hearts was closer to Holmes's view.[44] But he would trust the NASA administrator. Webb recalled having a conversation around this time in which he told the president that if they stuck together, things would work out all right, but if they disagreed, Webb could not guarantee that would be the case. The president assured him of his intent to "stick with" Webb.[45] He sent no formal response to Webb's letter.

At a NASA celebration that accompanied the end of the Mercury program in May 1963, Webb pointedly and publicly omitted Holmes from a long list of those he lauded for NASA's success in manned space flight. Livid, Holmes burst out to Seamans, "It's me or Jim." Seamans reported to Dryden and Webb that NASA could not get to the moon with Holmes and Webb at loggerheads, then went back to Holmes. "You said it yourself," he told him. "Webb and you can't live in the same house. And Webb's not about to leave." Holmes resigned in June. Under enormous pressure to hire a new manned space director, Webb called friends in industry. The name that surfaced was that of George E. Mueller (pronounced "Miller"), vice president of Space Technology Laboratories, a subsidiary of Thompson Ramo Wooldridge, Inc. After consulting with Dryden, Webb called Seamans, who was then away on vacation, to make sure he also thought well of Mueller.[46]

Joining NASA officially on September 1, 1963, the forty-five-year-old Mueller knew the program was the target of criticism and exacted from Webb the understanding that he would have a relatively free hand managing Apollo.[47] Virtually everyone agreed he was a first-rate engineer, and, as much a workaholic as Webb, he labored seven days a week and expected others to do so, scheduling important meetings on Sundays and holidays, calling the technical shots as he saw them, and seldom worrying if his decisions or manner of making them ruffled the feathers of subordinates.[48] He fastened on the lunar deadline and became its guardian.

Mueller profited from what had happened to Holmes. As determined as Holmes, he was circumspect with Webb and usually careful about what he said in reference to the administrator and to whom he said it. Bureaucratically adept, he found ways to get what he wanted without confronting those in superior positions. A man who neither sought nor received the attention Holmes did, he was content with the power his position afforded and quickly made a difference at NASA.

Shortly after he arrived, Mueller ordered an intense study made of where Apollo stood and found that Holmes had been right: Apollo was falling behind. Even with budgetary reprogramming and additional funds from the president and Congress, the program would have trouble making the Kennedy deadline. A lunar journey might not take place until 1971. Mueller thought he had a solution to the problem. As he saw it, NASA was overtesting. Von Braun's "German model" called for testing virtually every item connected with the rocket and spaceship separately and with painstaking detail, an approach with which Gilruth was not unsympathetic, since he worried constantly about the safety of the astronauts under his authority at Houston. The German model meant a long sequence of launches testing various parts of the Apollo configuration in space. But testing a number of the components together and launching them together as a complete system would eliminate many of the tests of individual stages and modules.[49]

Although von Braun and Gilruth opposed his approach as too risky, Mueller did not believe that simultaneous testing posed a much greater risk than testing separate components sequentially. Even if it did, how else to get to the moon by 1969? There had to be an "all-up" decision, as it came to be called, and Mueller held firm. This meant that the first flight of a *Saturn 5*, the moon rocket, would also be the first flight for various stages and modules now scheduled for separate tests. They would all be tested in flight at once.

When Webb decided to let Mueller proceed, the director informed a meeting of Houston and Huntsville officials of his all-up concept. Two days later, he sent a priority teletype message to relevant NASA groups, spelling out a new flight schedule, asking for responses in ten days and announcing his intention to post "an official schedule reflecting the philosophy outlined here" in three weeks. Webb strengthened Mueller's hand by a second reorganization of NASA, reversing the 1961 decision to put the centers under control of Seamans, already partially negated in 1962 for OMSF, as Webb sought to placate Holmes. Now, Webb made an across-the-board decision to shift power he had kept in the triad to program directors. Thus, Marshall, the Manned Space Center, and the Launch Operations Center (rechristened Kennedy Space Center in December) became solely Mueller's responsibility; Goddard Space Flight Center, the Jet Propulsion Laboratory, and certain other installations went to Newell, head of the Office of Space Science and Applications (OSSA); and the four former NACA labs—Langley, Ames, Lewis, and Flight—fell to Raymond Bisplinghoff, head of the Office of Advanced Research and Technology (OART). Whatever else it did, the reorganization sent a strong message to von Braun, Gilruth, and others that Webb was enlarging the influence of the program directors, now called associate administrators. In particular, it put Mueller in a good position to get his way if Gilruth and von Braun decided to fight the all-up decision. They chose not to do so, and the decision stood.[50]

In a meeting with Webb and Seamans, Mueller meanwhile said he needed more strong technical managers in headquarters, reporting to himself, to make Apollo a success. He produced a list of individuals, most of whom were in the air force. Seamans and Webb agreed, and, working primarily through a personal friend, General William F. "Bozo" McKee, deputy chief of staff of the air force, Webb maneuvered the necessary transfers.[51] General Samuel C. Phillips, one of the air force's top technical managers, was soon detailed to NASA as director of Apollo, under Mueller, and was later joined by twenty other able air force officers.[52] The crisis in Apollo leadership that had begun in 1962 with Holmes's mutiny thus ended in 1963 with an astute new manned space director, a stronger overall Apollo management team, and decisive steps to get Apollo back on schedule.

Clashing with McNamara

Another threat to Webb's control of the space program in the 1962–63 period came from Secretary of Defense Robert S. McNamara. Along with a basic institutional cooperation between DOD and NASA was an institutional rivalry—the air force wanted to be a manned space agency—as well as a personal competition between Webb and McNamara. They employed very different administrative styles toward a common end—control over their domain. The issue was the boundary of those domains, for both men had imperialistic tendencies.

In December 1962, McNamara began probing Webb to see how far he could go in establishing a DOD manned presence in space through Gemini. The air force argued that the near-earth activities Gemini would pioneer could have military significance in terms of manned reconnaissance, and McNamara agreed. He began by suggesting a basic division of labor: DOD would "take over all manned flights in earth orbit; NASA, all flights beyond earth orbit." This meant DOD would control space near earth, and NASA around the moon and beyond. Informally, McNamara proposed "a merger of NASA-DOD manned space programs under DOD management." When Webb objected, he formally proposed "a joint program."[53]

Webb saw the stakes as nothing short of NASA's independence as an agency. If McNamara took over Gemini, what would be next? Moreover, McNamara's proposal interfered with Apollo. "Gemini had been planned to meet the needs of NASA's manned program; it had been under way since December 1961." While NASA had designed Gemini with potential military applications "in mind," transferring Gemini might divert its being used as a building block for Apollo. "There were also compelling [international] political reasons for Gemini to remain within NASA, since the arrangement by which NASA operated tracking stations in Mexico, Nigeria, Zanzibar, and Spain prohibited their use to support military programs." If McNamara's terms were accepted, these stations "would be unavailable and the existence of a joint civilian-military program would jeopardize negotiations for tracking stations elsewhere. The effect of transferring Gemini to DOD would be to place in doubt NASA's image as a civilian agency dedicated to the peaceful exploration of space."[54]

Webb met with McNamara on the Gemini question in mid-January. McNamara's approach to negotiation, Webb said, was to "knock you down on the floor with a sledge-hammer, and then, while you're down, ask you to sign off on a particular decision." They had clashed before, and Webb felt personally slighted by the defense secretary's brusque style. He certainly had gone to some lengths to avoid dealing with McNamara, establishing various NASA-DOD liaison mechanisms to facilitate agreement short of the Webb-McNamara level. But the Gemini question (Air Force reportedly spoke of a "Blue Gemini" program under its auspices) was too important to be settled anywhere besides the top of the two agencies. In the meeting, Webb made it clear to McNamara that NASA control of Gemini was not negotiable. However, he would agree to Gemini's carrying certain military experiments. McNamara decided not to push Webb further, and the two men hammered out a pact. Under terms of the Gemini agreement, announced January 21, 1963, "Gemini was not to be thought of as a joint program, but rather as a program serving common needs, with the Department of Defense paying for the military features, NASA in full charge of the program, and the role of the [Gemini Program Planning] board [to plan experiments for NASA and DOD] strictly advisory." In addition, the agreement held that "DOD and NASA will initiate major new programs or projects in the field of manned space flight aimed chiefly at the attainment of experimental or other capabilities in near-earth orbit only by mutual agreement."[55]

That manned space was of primary interest to Webb and of secondary interest to McNamara gave Webb leverage in bargaining. If anything, McNamara was ambivalent about how far he wanted DOD (i.e., the air force) to move in this direction. He wanted both to extend his reach and to hold the air force in check. Webb understood this. NASA could not indefinitely keep DOD out of manned space, but Webb would try to limit and channel military ambitions so they did not hamper NASA's program. As Seamans put it, "McNamara was more powerful than Webb. But Webb had more guile."[56]

Challenges came as well from outside the executive branch. For two years after the Apollo decision Congress had been relatively docile. But in 1963, with Kennedy calling for a huge increase in NASA's budget, Congress decided to take a hard look at agency spending. In the midst of defending the NASA budget request, Webb found himself placed in a difficult position by Kennedy. The president, in a fall address to the UN General Assembly, declared that in the field of space there was room for U.S.-USSR "cooperation," for "joint efforts," including the possibility of a "joint expedition to the moon."

Kennedy had personally alerted Webb to his proposal and asked the NASA administrator to control his organization lest it undermine a possible foreign policy initiative. Webb had passed the word to his lieutenants

that they hold to the line of presidential policy—an action for which the president was grateful. However, the Kennedy speech offered ammunition to those who wanted to cut the NASA budget. The House had already agreed to cut the president's proposed budget by $600 million. Now, the Democratic chair of the House Appropriations Committee, Clarence Cannon of Missouri, decided to join with the Republicans on the Thomas subcommittee to cut NASA's appropriation an additional $900 million, which might have seriously "crippled" Apollo. Webb and Representative Thomas worked furiously behind the scenes to prevent the additional damage. The president contributed to their effort by stressing that a strong space program was essential to any cooperative endeavor. The prospect of a U.S.-USSR joint program eventually faded, and Webb and Thomas defeated the Cannon foray. In the end, the original request of $5.7 billion was cut to $5.1 billion, much more than the preceding year but less than Webb believed was needed. It was not clear who Congress was teaching a lesson, Webb or Kennedy.[57] What was clear was the necessity of having Thomas as an ally on the Appropriations Committee.

Two other disputes that came up in 1963 conveyed the message that Congress could no longer be taken for granted. The first concerned the final jewel in NASA's diadem of in-house facilities—the Electronics Research Center (ERC). Webb wanted this established in Cambridge, Massachusetts, near Harvard, MIT, and the Route 128 electronics complex, a key to his strategy of maximizing NASA's ability to create regional government-industry-university alliances for a stronger technological America. Many critics in Congress saw ERC as a payoff to supporters of President Kennedy and aid to his brother Edward, who had recently run for the Senate under the banner "I can do more for Massachusetts." The intense congressional controversy forced Webb to conduct a nationwide search for a site and to report in detail the rationale for his choice. In the end, ERC still went to Massachusetts.[58]

Similarly, the Sustaining University Program came under congressional fire in 1963. SUP had of necessity surfaced as a line item for the first time in the FY 1964 budget submission, for *Science* magazine had claimed that its level of spending made NASA the "biggest [agency] in graduate aid." Senator Gordon Allott, ranking Republican on the Appropriations subcommittee funding NASA, complained about millions of dollars going "into a program that no one had even heard of before." The Senate Appropriations Committee said it would not fund SUP unless NASA fully justified it. Special hearings by the Senate Space Committee had to be held on the program, and SUP was subsequently legitimated by Congress, two years after its birth.[59]

That Webb had to work hard to win his congressional battles in 1963 was partly related to the death in January of Senator Kerr, who was replaced as Space Committee chair by Senator Clinton P. Anderson, Democrat from New Mexico, a significant shift. During the Truman years, Anderson, as secretary of agriculture, had insisted that Webb, then budget director, come to *his* office to negotiate Agriculture's budget. Webb had refused, saying he worked for the president, not for Anderson. As a New Mexico senator, Anderson showed the same drive to control. Intellectually sharp and personally testy, he was perhaps the strongest senator on the Joint Committee on Atomic Energy, using that group to exercise unusually close influence over the Atomic Energy Commission. By the time he became chair of the Space Committee, however, he was in poor health, suffering from Parkinson's disease and other ailments that sapped his energy and made him unpredictable but did not change his will to control.

Various sources warned Webb that Anderson would try to find a way to bring him to heel.[60] Sure enough, Anderson challenged him on his reprogramming of funds already appropriated—particularly the shift of $400 million from Nova to Saturn. A hearing followed, during which Webb came under fire for excessive reprogramming and had his authority to shift funds subsequently narrowed. Afterward, Anderson walked up to him. "Well now, you've done all right," he told Webb. "You're going to get along just fine. We've known each other a long time, and we're going to get along just fine." Webb replied, "I'm sure glad to hear you say that, Senator, because several of my friends have said you told them that you were going to cut my throat, if you could. I didn't believe it, Senator, because we've been friends such a long time."[61]

If he did not stand up to the crusty senator right at the beginning, Webb realized, Anderson would push him into the ground, as he had done to some leaders of the Atomic Energy Commission. Although Webb was used to dealing with tough men who liked to exert power, Kerr among them, he and Kerr had been friends. Webb and Anderson never would be. In fact, Webb knew that, although Anderson supported NASA, he would have preferred someone else as administrator. He was just waiting for Webb to make the big mistake, Webb said later, so he could go for "the short hairs."[62]

Balancing Anderson to some extent was the strongest Republican on the Senate Space Committee, Margaret Chase Smith of Maine. The only

woman in the Senate, Smith had shown her remarkable blend of courage and independence when she became the first senator to actively speak out against Senator Joseph McCarthy in the 1950s. While staunchly conservative and critical of the Democrats, she voted her conscience. Webb went out of his way from the beginning to cultivate Senator Smith, using SUP to advantage, and they developed a strong mutual respect. She was important to the bipartisan image he wanted for NASA, and absolutely critical to bolstering the at-best-lukewarm support Webb got from Anderson.[63]

Using various strategies to win friends and maintain his authority, Webb neutralized opposition inside and outside NASA in these early years. Tested by the air force, the president's science adviser, his own Apollo director, astronauts, the secretary of defense, and Congress, he kept NASA on course—his course. As a result, by the end of 1963, NASA possessed not only the technical competence but the political clout to take America to the moon.

12

LEGACY

On May 25, 1961, with American pride and prestige at a low ebb, and the U.S. space program clearly behind that of the Soviet Union, President John F. Kennedy directed NASA to land a man on the moon and return him safely to Earth within the decade. The only human being to have flown into space at that point was Yuri A. Gagarin, a Soviet cosmonaut. The United States did not know precisely how it would get a man to the moon. NASA itself had known failure and ridicule more than success, and the U.S. Air Force was a stronger agency and an aggressive rival. The president's science adviser had earlier provided Kennedy with a report severely criticizing NASA's capability to lead the nation's space effort. In retrospect, the decision to go to the moon was not only bold, it was audacious.[1] When Apollo not only succeeded, but did so "on time and within budget," the man behind the agency—the man who built, maintained, defended, and drove NASA from 1961 to 1968—was James E. Webb.[2]

Webb's legacy is measured by Apollo, which stands today as a virtually unparalleled example of U.S. technological brilliance, as the mark of the moment when the human species took its "first journey to another world."[3] Further, in the broad sweep of history Apollo was a critical victory in the Cold War technological competition between the United States and Soviet Union. Certainly, subsequent history between the two nations would have differed greatly had Russians walked the moon, rather than Americans. Although many in the later 1960s did not believe that a race to the moon was under way, we now know that the Soviets had suffered a setback in their space program in the mid-1960s and then had regrouped and pushed for-

ward to try to overtake the United States. After America got to the moon first, and the Soviets experienced technical failures, the USSR abandoned its lunar project in the early 1970s.[4]

Apollo also stands as an outstanding example of public management. All the ballyhoo and hucksterism aside, the fact remains that a goal was proclaimed and accomplished. The goal was technically feasible, but what is technically feasible is not usually administratively possible in large-scale government programs. The Soviets could not get the job done. John E. Pike, director of space policy at the Federation of American Scientists, pointed out that "the reason we got to the moon before they did was that they had no one to pull this all together. The critical difference was, we out-managed them."[5]

As administrator of NASA during the 1960s, Webb deserves a large measure of the credit for pulling this all together, just as he would have suffered considerable damage to his reputation if the United States had failed. Even those who cared little for the goal, who, for example, wished the money could have gone to housing or education rather than space, ought to recognize the effectiveness of NASA's administration. To be sure, NASA had many able managers and, until the latter 1960s, plenty of money. But able managers, even with money, often fail. Webb sought a group power that he could raise to the highest level of performance possible. By melding government, industry, and university talent behind a monumental goal, he showed that a democracy could best an authoritarian regime in the development of big technology.

He was not perfect. The Apollo fire revealed that he himself had lost considerable control of the Office of Manned Space Flight, as well as of the principal contractor. He may not have been fair in his treatment then of certain officials in NASA and North American. Determined to manage the information flow, he was less than forthcoming with Congress and the media during the fire inquiry. His quest to reform higher education through SUP showed the limits of his administrative power and his naiveté about the way universities work. He failed to sell a firm post-Apollo program of sufficient scale to avoid retrenchment. At the same time, certainly, he was aggressive and protective, taking on all comers in defense of NASA. He believed in active government and revealed a technocratic zeal critics found dangerous. No wonder he is controversial with those who regard government-managed technology as a threat to democracy. Apollo gave him a stage on which to play a bigger-than-life role, and he performed with gusto.

Although Webb understood the dangers of administrative power, he believed it could be controlled and wielded in the public interest. While he made money in private enterprise, he was always public-service oriented, a bureaucratic entrepreneur, a man who believed he should try to improve society through government. In the name of Apollo, he propelled NASA to advance science and technology, enhance economic development, educate thousands of graduate students, and provide geographical balance in the distribution of scientific resources. Apollo was a means, as well as an end, to move a nation to a New Frontier, a Great Society, and preeminence.

There is debate not only about whether administrative leaders make a difference, but also—when they do—if this is desirable in a democratic government.[6] For Webb, effective government *required* administrative power. Bureaucracy was a fourth branch of government, the operational part of the American system. For democracy to work, bureaucracy had to function. Democracy was more than political speeches and elections; the people had to have confidence their government could carry out policies and programs well. An administrator not only needed resources and discretion to make government work and serve democracy, but also had to be responsive to president, Congress, and the public to get those resources and be granted the necessary autonomy to succeed. Webb felt that he had found the right balance between administrative discretion and democratic responsiveness in NASA, and thus the Holy Grail of public administration. Afterward, he worried that America had pulled back from not only the new frontier of space but also the frontier of management demonstrated in getting to the moon. For Webb, it was not Apollo, but NASA as a model of administrative excellence that was his legacy. He truly believed that if a nation could put a man on the moon, it could manage its other large public problems.

Whether or not Webb had found a Holy Grail of management is conjectural. The triad, scouts, formal and informal chains of command, checks and balances within NASA and between NASA and its contractors, participation, supervision—he used these and other management schemes pragmatically to create an organizational paragon. For the most part they seemed to work, but perhaps less because of their inherent quality than Webb's ability to stay in charge and influence events. Management innovation was both a reality and public relations strategy for Webb. Above all, he understood the importance of power: its nurture, use, and loss. With this understanding he showed what a public executive with the right mix of energy, conviction, skill, wide-ranging contacts, and experience can do when conditions are ripe. He was cunning, guileful, manipulative, and hyperbolic—a lot like Lyndon Johnson. If he did not cross the line of administrative accountability, he surely edged up against it. But unlike Johnson, whose orientation was to legislation, Webb cared deeply about the effective administration of public affairs and was able to inspire those under him to new heights. He pushed a vast army of specialists in a common direction, at breakneck speed. A bold risk taker, he confronted pressures from astronauts, the president's science adviser, the air force, the defense secretary, other NASA executives, Congress, media, industry, universities, and many others, and did not flinch. Rather than break under political pressure, he used rhetorical and coalition-building techniques to impose pressures and coopt others. Organizing and reorganizing NASA as circumstances changed, using the resources at his disposal to reward and punish, he kept reign on his internal forces while orchestrating external support. Throughout, he focused on Apollo's goal and what it took to get there. If he ever questioned that goal or the country's capacity to attain it, he did not communicate such doubts to others.

Containing the Apollo fire crisis and moving NASA to recovery expended much of his own personal political capital. Still, he could not have achieved what he did as an administrator without also having been an extraordinary politician. Nor could he have been successful as a politician had he not performed well as a gifted administrator. This dual capacity sets Webb apart from the legion of others who run agencies in Washington. He could play the external and internal roles of public executive and make one strengthen the other. NASA bore his mark for many years and was synonymous with technological success up until the *Challenger* disaster of 1986, when the decline of this great organization became evident.

Webb traveled far from rural North Carolina to Washington, from the Bureau of the Budget to the State Department to NASA. A good match with the Bureau of the Budget, less so with the State Department, in NASA in the 1960s Webb had a chance to shine. Whatever his drawbacks, this formidable, dedicated, and intensely driven man used his mastery of the governmental process to lead the United States to the moon. He fought the battles in Washington, overcame bureaucratic, legislative, and industrial barriers in NASA's way, and shielded his agency when it counted most. More than a political operator or large-scale manager, Webb embodied the triumph and ambiguity of administrative leadership in America. At NASA, where he tenaciously and forcefully pursued a spectacular vision of the national interest, he indeed was the power behind Apollo.

Notes

Introduction

1. John F. Kennedy, "Message to Congress, May 25, 1961," *Public Papers of the Presidents of the United States, January 20–December 31, 1961* (Washington: USGPO, 1962), 404.

2. John Noble Wilford, *We Reach the Moon* (New York: Bantam Books, 1969), 269.

3. Bruce Lambert, "James Webb, Who Led Moon Program, Dies at 85," *New York Times*, March 29, 1992; Richard Pearson, "James E. Webb Dies at 85; Was NASA Chief in 1960s," *Washington Post*, March 29, 1992.

4. It has been calculated that $20 billion in 1961 dollars would be $94 billion in 1990 dollars. See *Report of the Advisory Committee on the Future of the U.S. Space Program* (Washington: USGPO, 1990), 11–13. The exact figures are problematic, owing to disagreements about what should or should not be included in Apollo over the course of the program's evolution since 1961. Moreover, it is always difficult to determine the "value" of money from one decade to the next. Nevertheless, it is worth trying to give some impression of the magnitude of this endeavor. Like the Manhattan Project, Apollo stands out from other R&D endeavors in history in terms of sheer scale.

5. See Herbert Kaufman, *Time, Change, and Organizations: Natural Selection in a Perilous Environment* (Chatham, N.J.: Chatham House, 1985), and *The Administrative Behavior of Federal Bureau Chiefs* (Washington: Brookings Institution, 1981).

6. Thomas Carlyle, *On Heroes, Hero-Worship, and the Heroic in History* (1840), cited in Jameson W. Doig and Erwin C. Hargrove, eds., *Leadership and Innovation: Entrepreneurs in Government* (Baltimore: Johns Hopkins University Press, 1990), 1.

7. Fred I. Greenstein, *The Hidden-Hand Presidency: Eisenhower as Leader* (New

York: Basic, 1982); on general leadership theory, see Doig and Hargrove, *Leadership and Innovation*, 1–22.

8. James MacGregor Burns, *Leadership* (New York: Harper and Row, 1978), 4.

9. John M. Logsdon, quoted in Douglas Isbell, "James Webb, Former NASA Administrator, Dies at 85," *Space News*, April 6–12, 1992, 26; Obituary, *Aviation Week and Space Technology*, April 6, 1992, 15.

10. Clayton R. Koppes, *JPL and the American Space Program* (New Haven: Yale University Press, 1982), 143.

11. Tom Wolfe, *The Right Stuff* (New York: Bantam, 1983), 227.

12. John Mecklin, "Jim Webb's Earthy Management of Space," *Fortune*, August 1967, 87.

13. James E. Webb, *Space Age Management: The Large-Scale Approach* (New York: McGraw-Hill, 1969).

14. Mecklin, "Jim Webb's Earthy Management of Space," 83.

15. Robert L. Haught, ed., *Giants in Management* (Washington: National Academy of Public Administration, 1985), 21–34; Doig and Hargrove, *Leadership and Innovation*, 139–68.

16. Leonard Sayles, "The Unsung Profession," *Issues and Observations*, May 1984, 1.

17. Charles Sheffield, review of *Liftoff: The Story of America's Adventure in Space*, by Michael Collins, *Washington Post Book World*, June 19, 1988.

18. Wayne Biddle, "Two Faces of Catastrophe," *Air and Space*, August/September 1990, 46–49.

19. Mecklin, "Jim Webb's Earthy Management of Space," 87, 84.

20. Leonard R. Sayles and Margaret K. Chandler, *Managing Large Systems: Organizations for the Future* (New York: Harper and Row, 1971); Walter A. McDougall, *. . . the Heavens and the Earth: A Political History of the Space Age* (New York: Basic, 1985), chap. 18, "Big Operator: James Webb's Space Age America." McDougall refers to "Webb and the American technocrats" (388) and to Webb's "technocratic vision" (387); it is clear he does not share or approve of that vision.

21. Walter A. McDougall, "Big Operator," 362–73.

22. Mecklin, "Jim Webb's Earthy Management of Space," 84.

23. James M. Beggs, who worked for Webb and later became NASA administrator himself, said Webb aimed high—to create "the perfect management system." See his "James E. Webb: A Force for Excellence," the Inaugural Lecture of the Fund for Excellence in Public Administration (Washington: National Academy of Public Administration, 1983), 12; Biddle, "Two Faces of Catastrophe," 47.

24. Norton E. Long, "Power and Administration," in *Bureaucratic Power in National Policymaking*, ed. Francis E. Rourke, 4th ed. (Boston: Little, Brown, 1986), 7.

25. Wolfe, *The Right Stuff*, 226–27.

Chapter 1. The Making of a Public Executive

1. James Webb, interview by David DeVorkin and Joseph Tatarewicz, February 22, 1985, National Air and Space Museum, Washington (hereafter NASM); James Webb, interview by Robert Sherrod, April 28, 1971, NASA History Division, Reference Collection, Washington.

2. Olive Webb Wharton, interviews by Bailey Webb, August 1986-May 1, 1990, Durham, N.C.

3. Arthur S. Link and Richard L. McCormick, *Progressivism* (Arlington Heights, Ill.: Harlan Davidson, 1983), 90.

4. Woodrow Wilson, as president of Princeton, called the Bell Buckle School one of the best in the country. See Edmund Fuller, "Old Sawney's Inspiring School," *Wall Street Journal*, September 20, 1971.

5. "This Is Epochal History Making Day for Granville County," editorial, *Oxford (N.C.) Public Ledger*, March 6, 1964. The editorial praised J.F. Webb upon the dedication of a high school building in his name and said that "no man in the history of Granville County has accomplished more in public education than Mr. Webb."

6. Webb interview, April 28, 1971.

7. Fuller, "Old Sawney's Inspiring School"; Wharton interviews, August 1986-May 1, 1990.

8. Fred Webb, interview by author, June 28, 1990, Oxford, N.C.; also see Fuller, "Old Sawney's Inspiring School"; Bailey Webb, "The Family Relationships of James Edwin Webb," personal document provided the author. "Sawney was always looming in the background of the Webb family," Olive Webb Wharton said in our interview on June 28, 1990, in Durham, N.C. Matters of integrity were driven home in the Webb household, often by quoting Sawney Webb, under whom J.F. had studied. Fred Webb himself quoted Sawney—a quote he got from J.F.: "Shortly before death, when you come to the end, you realize that nothing is more important than character and what you have done for other people." Said Fred: "This is the sort of thing we always heard from father. The family really thought about this." Fred Webb interview, June 28, 1990.

9. Webb interviews, February 22, 1985, April 28, 1971; Wharton interview, June 28, 1990.

10. Wharton interviews, June 28, 1990, August 1986-May 1990.

11. Webb interviews, February 22, 1985, April 28, 1971.

12. Wharton interviews, August 1986-May 1990, June 28, 1990.

13. Webb interviews, February 22, 1985, April 28, 1971; Wharton interviews, August 1986-May 1990.

14. Link and McCormick, *Progressivism*, 85-96.

15. Wharton interviews, August 1986-May 1990.

16. Webb interviews, February 22, 1985, April 28, 1971.

17. Wharton interviews, August 1986-May 1990, June 28, 1990.

gress (by which time the Webb-McNamara memo had gone to President Kennedy and been approved), Webb and Patsy went to a dinner at the Carnegie Institution in Washington. Webb's longtime friend Vannevar Bush saw Webb come in the door. Webb wrote: He "was so incensed at the furor [of the Shepard events] that he grabbed me by my lapels to shake me—accusing me of 'exhaltation' of a very dangerous public mood." Webb, Handwritten note, January 27, 1975, as cover to letter from Webb to Vannevar Bush, May 15, 1961, Webb Papers, Truman Library. He wrote another Apollo critic: "I am certain we must increase the capacity of our American scientific community and particularly our universities to back up our program." Webb, letter to Jerome B. Wiesner, May 15, 1961, Webb Papers, Truman Library.

43. Walter A. McDougall, . . . *the Heavens and the Earth: A Political History of the Space Age* (New York: Basic, 1985), 361.

44. James Webb, memo to Vice President, May 23, 1961, NASA History Office.

45. Ibid.

46. John F. Kennedy, Message to Congress, May 25, 1961, *Public Papers of the Presidents of the United States, January 20–December 31, 1961* (Washington: USGPO, 1962).

47. Seamans, "Voyage to the Moon."

48. Ibid.

Chapter 6. Launching a Stronger NASA

1. James Webb, interview by David DeVorkin, Allan Needell, and Joseph Tatarewicz, April 12, 1985, National Air and Space Museum, Washington (hereafter NASM).

2. James Webb, interview by author, April 12, 1985, Washington; James Webb, interview by David DeVorkin, Joseph Tatarewicz, and Linda Ezell, March 22, 1985, NASM.

3. James Webb, interview by Martin Collins, October 15, 1985, NASM.

4. James Webb, interview by Allan Needell and Martin Collins, September 10, 1985, NASM.

5. James Webb, interview by author and Martin Collins, October 3, 1986, NASM.

6. In August 1961, Webb wrote T. Keith Glennan, his predecessor as NASA administrator, that he doubted the political honeymoon following Kennedy's decision would last long. Meanwhile: "I am finding, with some unhappiness, that I am expected to be an 'image' of an advancing, ongoing program." Although he had tried to "stay in the background," the media were insisting on personality sketches and interviews. Webb was indeed more organization man than publicity seeker. But he was finding that the nature of space precluded his being invisible. He sought to use this potential prominence to NASA's advantage by presenting an optimistic and can-do image. Webb, letter to T. Keith Glennan, August 22, 1961, Webb Papers, Truman Library. He noted to a friend from Oklahoma: "This is a really tough job—with the out-in-front technological requirements, and the eyes of all the world on every move you make." Webb, letter to Edward T. Gaylord, August 24, 1961, Webb Papers, Truman Library.

7. Webb interview, October 3, 1986.

8. Ibid.

9. R. P. Young, interview by author, August 1, 1990, Washington.

10. Webb, memo for Mr. Siepert, "How to make the Dryden-Seamans-Webb group more effective on important decisions," August 28, 1961, Webb Papers, Truman Library.

11. Webb told Cox how important the president was for getting NASA's message out. The point was to show the president how the space program was helping him with his objectives. First, NASA should be "feeding" information to the president, and, second, "his words" would carry the NASA message. Webb mentioned other cabinet members who could help NASA if NASA provided them with the appropriate information. Webb, memo for Dr. Cox, December 4, 1961, Webb Papers, Truman Library.

12. Webb interview, October 15, 1985; Arthur Raymond, interview by author, November 30, 1990, Brentwood, Calif.

13. Raymond was brought aboard by Mervin Kelly, retired executive with AT&T. Along with some others they formed an informal kitchen cabinet to Webb, commenting on organizational issues and personnel problems. Kelly left after a few years. Raymond was truly used as a scout and gave Webb insights into goings on in his centers and industry—one more set of eyes and ears. Mervin Kelly, letter to Arthur Raymond, June 30, 1961, provided author from Raymond's personal files; Webb, letter to Arthur Raymond, August 1, 1961, Webb Papers, Truman Library; Webb, letter to Mervin Kelly, January 7, 1963, NASA History Office; Webb, letter to Arthur Raymond, August 12, 1963, and Raymond letter to Webb, August 14, 1963, both provided author from Raymond's personal files. Playing a different role as an informal confidant was Webb's friend Berkner, who "thought big" and shared Webb's enthusiasm for harnessing large-scale, interdisciplinary science to national problems; the two men fueled one another's visions for years. Berkner died in 1967. See Allan A. Needell, "From Military Research to Big Science: Lloyd Berkner and Science Statesmanship in the Postwar Era," in *Big Science: The Growth of Large-Scale Research*, ed. Peter Galison and Bruce Hevly (Palo Alto, Calif.: Stanford University Press, 1992), 290–311.

14. Webb interview, March 15, 1985.

15. Ibid.

16. Robert Seamans, interview by author, November 16, 1990, Cambridge, Mass.; see also Courtney G. Brooks, James M. Grimwood, and Loyd S. Swenson, Jr., *Chariots for Apollo: A History of Manned Lunar Spacecraft* (Washington: NASA, 1979), 41; Webb, letter to C. W. Perella, September 9, 1961, NASA History Office.

17. Kenneth O'Donnell, interview by Robert Sherrod, May 13, 1971, NASA History Office; Webb, memo for the president, September 14, 1961, Webb Papers, Truman Library. The memo to Kennedy came on the heels of an acrimonious telephone conversation between Webb and Gov. John Volpe of Massachusetts, who had been pressuring Webb to site the Manned Spacecraft Center in his state. Webb ended the conversation by saying that if Volpe did not watch out, Webb would be "issuing a statement to the press," and Volpe "probably wouldn't like it." Webb, memo of telephone conversation, September 12, 1961, Webb Papers, Truman Library.

18. Robert Gilruth, interview by David DeVorkin, March 2, 1987, NASM. Webb made sure Senator Byrd did not think too badly of him, informing Byrd of a $6 million expansion of the Langley Field installation. Webb, letter to Senator Harry Byrd, October 24, 1961, Webb Papers, Truman Library.

19. Charles Murray and Catherine Bly Cox, *Apollo: The Race to the Moon* (New York: Simon and Schuster, 1989), 168.

20. Ibid., 170; James Webb, interview by author, January 30, 1992, Washington. Not long thereafter, NASA and North American ran into disagreements, and by January 1963 Webb was even saying that NASA might have to "seek another contractor, even though the penalties would be very great." Webb, memo for Dr. Seamans, January 31, 1963, NASA History Office.

21. Seamans interview, November 16, 1990. The possibility of Admiral Raborn's becoming Apollo czar got into the papers, and this gave the navy time to act. Webb insisted he was "not necessarily looking for a czar but rather for a form of organization that would get our total job done." Webb, memo for Dr. Seamans, August 3, 1961, Webb Papers, Truman Library.

22. Robert Rosholt, *An Administrative History of NASA, 1958–1963* (Washington: NASA, 1966), 224–25.

23. Joseph Trento, *Prescription for Disaster* (New York: Crown, 1987), 51; see also Barton C. Hacker and James M. Grimwood, *On the Shoulders of Titans: A History of Project Gemini* (Washington: NASA, 1977).

24. Trento, *Prescription for Disaster*, 52.

25. Webb's interest in the "image aspects" extended to the name. As Gemini became the likely name, Webb asked his public affairs director to "check carefully to see that 'Gemini' does not have any unfortunate connotation which would cause trouble in some parts of the world, in mythology, or in some other area that may relate to public affairs." Webb, memo for Hiden Cox, December 19, 1961, Webb Papers, Truman Library.

26. John Simpson, interview by author, August 1, 1991, Chicago.

27. Webb, letter to Lee DuBridge, August 29, 1961, Webb Papers, Truman Library.

28. Webb interview, April 12, 1985.

29. W. Henry Lambright, *Launching NASA's Sustaining University Program* (Syracuse, N.Y.: InterUniversity Case Program, 1969), VII-2; Webb, letter to David Bell, December 6, 1961, Webb Papers, Truman Library.

30. Murray and Cox, *Apollo*, 139.

31. Ibid., 141, 143; Jerome Wiesner, interview by author, November 15, 1990, Cambridge, Mass. According to Howard E. McCurdy, after the public dispute with Wiesner Webb stood up in the front of the bus that was taking NASA and some other government officials to their next stop and declared, "Hey, we're running this show and we're going to run it." *Inside NASA: High Technology and Organizational Change in the U.S. Space Program* (Baltimore: Johns Hopkins University Press, 1993), 88.

32. John Noble Wilford, *We Reach the Moon* (New York: Bantam, 1969), 48; Wiesner interview, November 15, 1990; Seamans interview, November 16, 1990. As late as October 29, 1962, Wiesner was trying to get Webb to give him proposals from contractors associated with LOR techniques. Webb, memo for Dr. Seamans, October 29, 1962, Webb Papers, Truman Library.

33. James Webb, interview by Martin Collins and Allan Needell, November 4, 1985, NASM.

34. Hiden T. Cox, interview by author, April 10, 1990, Long Beach, Calif.; Robert Sherrod, "The Selling of the Astronauts," *Columbia Journalism Review*, May/June 1973, 16–25; Webb, memo for Robert Gilruth and Hiden Cox, March 16, 1962, and memo to Hiden Cox, March 30, 1962, Webb Papers, Truman Library. Webb also had to set guidelines for Wernher von Braun, whose outside speaking engagements (for large fees) had become excessive, in Webb's view. See Webb, memo for Wernher von Braun, December 5, 1961, Webb Papers, Truman Library.

35. Webb interview, October 3, 1986.

36. James Webb, interview by author, March 26–27, 1991, Washington.

37. Robert Seamans, interview by author, December 20, 1990, Cambridge, Mass.

38. "Reaching for the Moon," *Time*, August 10, 1962, 52–57.

39. Wilford, *We Reach the Moon*, 64; Murray and Cox, *Apollo*, 152.

40. R. P. Young, memo for the record, November 20, 1962, and see also P. T. Drotning, memos to Col. Young, November 19, 20, 1962, Webb Papers, Truman Library.

41. "Space in Earthly Trouble," *Time* 80, November 23, 1962, 15.

42. Seamans interview, November 16, 1990.

43. Webb, letter to the president, November 30, 1962, Webb Papers, Truman Library.

44. Wiesner interview, November 15, 1990.

45. Webb interview, October 15, 1985.

46. Seamans interview, November 16, 1990. Webb arranged for Holmes to go back to his former company at a salary larger than he had when he came to NASA. Holmes, however, chose to go to a new company, Raytheon, and later became its president.

47. George Mueller, interview by author, September 21, 1990, Washington.

48. Murray and Cox, *Apollo*, 158–60.

49. Ibid., 156.

50. Ibid., 160, 162.

51. Seamans interview, December 20, 1990. Webb also dealt with Secretary of Defense Robert S. McNamara and Air Force Secretary Eugene M. Zuckert. Webb, memo for Dr. Mueller, January 14, 1964, NASA History Office; George Mueller, memo for the administrator, "Utilization of Air Force Program Management Personnel," September 26, 1963, Webb Papers, Truman Library.

52. Brooks, Grimwood, and Swenson, *Chariots for Apollo*, 128.

53. Arnold Levine, *Managing NASA in the Apollo Era* (Washington: NASA, 1982), 230.

54. Ibid., 231.

55. Ibid., 231, 321; Webb, memo for Dr. Seamans, January 10, 1963, Webb, letter to Robert McNamara, January 16, 1963, McNamara, letter to Webb, September 16, 1963, and Webb, letter to McNamara, September 23, 1963, NASA History Office. Webb's characterization of McNamara's negotiation style is from my interview with Webb, January 8, 1991, Washington.

56. If Webb was uncomfortable with McNamara, McNamara was uncomfortable with Webb. The defense secretary liked meetings short and to the point. Webb talked too much for him and was too "political." Robert McNamara, interview by author, September 26, 1991, Washington. In the early period after the Apollo decision, Webb and McNamara met regularly for lunches, accompanied by aides, to facilitate coordination. At one of these lunches, McNamara lectured Webb, so offending the NASA administrator that he and Seamans walked out, and the regular lunches were discontinued. Although the two senior officials dealt with one another as little as possible thereafter, they had to cooperate to some extent for common interests. Webb used Seamans as a surrogate, and McNamara used similarly appropriate substitutes. Seamans interview, December 20, 1990. There are only a few references on space in Deborah Shapley's biography of McNamara, *Promise and Power: The Life and Times of Robert McNamara* (Boston: Little, Brown, 1993), indicating a limited interest on McNamara's part in this field compared to his other responsibilities.

57. Rosholt, *An Administrative History of NASA*, 288, 289.

58. For a case history of the Electronics Research Center decision, see Thomas Murphy, ed., *Science, Geopolitics, and Federal Spending* (Lexington, Mass.: Heath, Lexington, 1971), 225–64; Webb interview, October 15, 1985. A particular critic of the ERC decision was Joseph Karth, Democrat from Minnesota, an important member of the House Space Committee. Webb, memo for Dr. Seamans, March 20, 1962, NASA History Office. But others on the space committee also rebelled against the decision, forcing Webb to go through a new site-selection process. Webb, memo for Dr. Dryden, August 13, 1963, NASA History Office.

59. Daniel S. Greenberg, "NASA: New Fellowship Program Will Make Space Agency Biggest in Graduate Aid," *Science*, January 4, 1963, 23–24; Lambright, *Launching NASA's Sustaining University Program*, X-17, X-21.

60. One of Webb's sources was Earl "Red" H. Blaik, an executive with Avco Corporation and former West Point football coach, who warned Webb that Anderson could be "a first class obstructionist." Blaik, letter to Webb, May 17, 1963, James Webb personal files, Washington.

61. Webb interview, October 15, 1985. Reprogramming was discussed during hearings relating to the NASA authorization for fiscal year 1964. Senate Committee on Aeronautical and Space Sciences, *NASA Authorization for FY 1964* (Washington: USGPO, 1963).

62. Webb interview, October 15, 1985. Anderson frequently made Webb aware that he was watching. In October 1964, he commented to Webb somewhat casually, "I see you've issued some new patent regulations." Webb said yes, and these had been discussed with the committee and they were in accord with the policy of the White House. Anderson responded that he wondered if they were in accord with the policy laid down in the law. Webb kept stretching his authority, and Anderson kept trying to pull him in. Webb, memo for Mr. Callaghan, October 12, 1964, NASA History Office. What Webb had to help keep Anderson friendly was power to promote and support projects dear to Anderson's heart, including the NERVA nuclear rocket, a joint development effort of NASA and AEC, drawing on work by facilities in New Mexico. This long-term, long-range project was quite vulnerable to shifts in political fortune and huge in scale, well over a billion dollars for research and development. So Anderson needed Webb, much as Webb needed Anderson, and Webb did in fact try to take care of NERVA. It was folded into what became a package of post-Apollo activities, but its fate ultimately turned on the fate of NASA after Apollo, and it was canceled in 1972. Rosholt, *An Administrative History of NASA*, 254–55. W. Henry Lambright, *Governing Science and Technology* (New York: Oxford University Press, 1976), 124–27.

63. Webb interview, October 15, 1985; Margaret Chase Smith, interview by author, September 28, 1991, Hobart College, Geneva, N.Y.; Webb, memo for Paul Dembling, August 15, 1961; Webb, letter to Margaret Chase Smith, August 18, 1961, Webb, memo to Hugh Dryden, August 21, 1961, and Webb, letter to Margaret Chase Smith, April 23, 1962, Webb Papers, Truman Library; Webb, memo for the president, subject NASA activities in Maine, October 17, 1963, NASA History Office. This last memo details Webb's efforts to win Smith's support. Webb found she was especially interested in his Frontiers of Science Foundation experience, and she in fact took a delegation of Maine leaders to Oklahoma to see what had taken place there.

Chapter 7. The Struggle to Maintain Momentum

1. Lyndon B. Johnson, *The Vantage Point* (New York: Popular Library, 1971), 279.

2. Thomas Murphy, interview by author, December 21, 1990, Washington.

3. Robert B. Young, interview by Robert Sherrod, January 12, 1972, NASA History Office; "Huntsville Visit Scheduled by NASA Administrator Webb," *Marshall [Al-*

Robert Dallek, "Johnson, Project Apollo, and the Politics of Space Program Planning," paper presented at NASA Conference on Presidential Leadership, Congress, and the U.S. Space Program, Washington, March 1993; also, Lambright, *Presidential Management of Science and Technology*, 149–50.

41. Dallek, "Johnson, Project Apollo."

42. James M. Beggs, "James E. Webb: A Force for Excellence," Inaugural Lecture of the Fund for Excellence in Public Administration (Washington: NAPA, 1984), 16.

43. Paine interview, April 6, 1990.

44. "National Aeronautics and Space Administration Awards," *Weekly Compilation of Presidential Documents* (week ending November 8, 1968) (Washington: USGPO, 1968), 1569. It is noteworthy that Webb's frequent critic, Senator Anderson, showed his respect by inserting praise for Webb in the *Congressional Record* at this time and also remarked: "He probably never will receive all the credit that is due him." *Congressional Record*, Senate, 90th Cong., 2d Sess., 1968, vol 114, pt. 24: 30875.

45. John Noble Wilford, *We Reach the Moon* (New York: Bantam, 1969), 205.

Chapter 11. The Moon and After

1. Thomas Paine, interview by author, April 6, 1990, Los Angeles.

2. Ibid.

3. Webb, letter to Eugene Emme, January 16, 1969, NASA History Office; James Webb, interview by author, December 12, 1990, Washington.

4. Robert C. Seamans, "Voyage to the Moon: A View from NASA Headquarters." See chap. 5, n. 38, in this work.

5. John Noble Wilford, *We Reach the Moon* (New York: Bantam, 1969), 257–60.

6. Ibid., 269.

7. Craig Covault, "Soviet Union Reveals Moon Rocket Design That Failed to Beat U.S. to Lunar Landing," *Aviation Week and Space Technology*, February 18, 1991, 58–59.

8. Joseph Trento, *Prescription for Disaster* (New York: Crown, 1987), 88.

9. Charles Murray and Catherine Bly Cox, *Apollo: The Race to the Moon* (New York: Simon and Schuster, 1989), 449. The Nixon administration also terminated DOD's Manned Orbital Laboratory.

10. James E. Webb, *Space Age Management: The Large-Scale Approach* (New York: McGraw-Hill, 1969), 15.

11. Ibid., 17, 27.

12. Al Neuharth, *Confessions of an S.O.B.* (New York: Penguin, 1992), 76, 91–93.

13. Quoted in Elmer B. Staats, "James E. Webb, Space Age Manager," in *Giants in Management*, ed. Robert L. Haught (Washington: NAPA, 1985), 29.

14. Alan Ullberg, interview by author, October 26, 1990, Washington.

15. Staats, "James E. Webb," 30.

16. Webb, letter to Bailey Webb, August 15, 1975, James Webb Personal Files, Washington.

17. Christian Williams, "James Webb and NASA's Reach for the Moon," *Washington Post*, September 24, 1981.

18. President George Bush, letter to Mrs. Webb, April 7, 1992, James Webb Personal Files, Washington.

Chapter 12. Legacy

1. John Noble Wilford, *We Reach the Moon* (New York: Bantam, 1969), 68.

2. *Report of the Advisory Committee on the Future of the U.S. Space Program* (Washington: USGPO, 1990), 13. Thanks to his "administrative discount on technical optimism," Webb was realistic about Apollo cost and schedule projections. He could also get political acceptance of realistic cost estimates in 1961. For a discussion of "optimism" and "realism" in estimating big technology programs, and their importance in selling programs and conveying the image as well as the reality of good management, see W. Henry Lambright, *Governing Science and Technology* (New York: Oxford University Press, 1976), 53–56; see also Harvey Sapolsky, *The Polaris System Developmment: Bureaucratic and Programmatic Success in Government* (Cambridge: Harvard University Press, 1972).

3. Charles Murray and Catherine Bly Cox, *Apollo: The Race to the Moon* (New York: Simon and Schuster, 1989), 459.

4. Craig Covault, "Soviet Union Reveals Moon Rocket Design That Failed to Beat U.S. to Lunar Landing," *Aviation Week and Space Technology*, February 18, 1991, 58–59; see also John M. Logsdon and Alain Dupas, "Was the Race to the Moon Real?" *Scientific American*, vol. 270, no. 6, June 1994, 36–43.

5. Quoted in Bruce Lambert, "James Webb, Who Led Moon Program, Dies at 85," *New York Times*, March 29, 1992.

6. Jameson W. Doig and Erwin C. Hargrove, eds., *Leadership and Innovation: Entrepreneurs in Government* (Baltimore: Johns Hopkins University Press, 1990); see also Larry Terry, "Why We Should Abandon the Misconceived Quest to Reconcile Public Entrepreneurship with Democracy," and Carl J. Bellone and George Frederick Goerl, "In Defense of Civic-Regarding Entrepreneurship or Helping Wolves to Promote Good Citizenship," *Public Administration Review* 53 (July/Aug. 1993): 393–95 and 396–98.